KOR STRUCTURAL · WOOD FRAME · OCTOBER 2026

Wood frame: one standard, modelled

We compared how three wood-frame models were drafted, line by line. Every one does it differently, and the drawings pay for it. This is what we found, why it happens, and how one standard — held in the Standard Details catalogue and placed by the Virtual Drafter — fixes it.

WHAT WE COMPARED

Two recent jobs and the office wood template, each read element by element from the Revit model: **Sherwood North Rd (31207-01)**, **Edgar, 199 St (31165-01)**, and the **office wood master** in 00 Templates. All counts below are from those models, measured on 25 October 2026.

THREE MODELS, THREE WAYS

	SHERWOOD	EDGAR	OFFICE WOOD MASTER
Joists	Not modelled — drawn as 212 span symbols	Modelled: 5,459 joist members, plus span symbols	Drawn in 2D as detail items
Beams	911 beams, all one placeholder type; the mark carries the size	2,085 beams at real sections — but in 7 families and 15 types	Drawn in 2D on the plan; the model sits in a linked file (not reviewed here)
Hold-downs	One family (## - HD)	Two families for the same thing (## - HD 114, KOR_HD 12)	Drawn in 2D as detail items
Framing drawn in	The plan of the level above	A ceiling plan of the level	The plan of the level itself
Typical floors	Drawn once for levels 3–5	Every level drawn, even where identical	Every level drawn
Mark text	Colour-coded types (pink, red) that all print black	One plain type (T1), bar a few opaque variants	Mixed — a looped style on 459 of 763 beam marks
Schedules on sheet	One combined legend	Seven separate legends	Seven separate legends

WHAT IT COSTS

177 beams

On one job, the model and the drawings disagree.

Of 607 beam marks on Edgar's framing plans, 564 sit on a beam of the right size — but 197 sit on the wrong *product*. 177 are marked B23 (Timberstrand) on beams modelled as Parallam of the same size. The marks followed a substitution; the model did not. Anything taken off the model carries the old member.

7 families

For what are mostly the same members.

Edgar's beams use KOR_Beam Lumber, KOR_Beam Lumber1, KOR - Beam Lumber, ##_KOR - Beam Lumber, KOR_Beam Engineered Wood, ## - Beam Engineered Wood and a copy of it. Duplicates like these typically come from loading a family again from another job, and schedules, tags and quantities can split across them.

No standard exists to follow.

The office standards template carries no wood beam or joist framing family. The Standard Details catalogue's wood palette holds drafting symbols, not modelled beam or joist types. With nothing standard to load, every job builds its own.

Models drift in ways nobody sees.

On Edgar, two of twelve levels use a level type measured from a different base point — 93 ft apart — and 2,366 of 7,544 framing members hang off Level 1 with offsets of whole storeys. It looks right on the sheet; anything that reads level heights or member levels from the model gets it wrong.

WHY IT HAPPENS

Until recently the firm was told wood frame could not be done in Revit. The office wood master shows it: the framing is drawn in 2D on top of a concrete-only model. When jobs began modelling wood, there was no agreed family set, no agreed way to place it, and nothing to check against — so each drafter and each engineer made the convenient choice for that job. None of the differences above is a mistake on its own. Together they mean no two wood jobs are drawn alike.

HOW IT GETS FIXED

1 · MODELLED

Wood is modelled

Beams at their real sections, joists as members, posts, hold-downs and walls as elements. On a framing plan, 2D is annotation only — marks, span symbols, tags. Standard details (sections, connections) are 2D by nature and stay as they are.

DECIDED

2 · ONE FAMILY SET

One standard family set

The wood model families and types go into the Standard Details catalogue — one beam type per master-schedule entry, size *and* product. Reviewed and approved like every standard detail. Duplicates and job copies are retired, not loaded.

BEING SET UP

3 · DRAWN FROM IT

The Virtual Drafter draws from it

It places only standard types, and takes each beam's type from its mark through the master schedule. It never copies a job's own families — so what it produces is clean by construction.

IN PROGRESS

4 · DECISIONS KEPT

Each job's choices are recorded

How a job is drafted is recorded once, in the firm's standards database, and confirmed by its drafter or engineer — never re-decided by whoever opens the file next.

RUNNING

5 · CHECKED

Model and drawings checked against each other

An automatic check pairs every beam mark with its beam and compares it with the schedule. It found the 177 above in seconds, and can run on every job before issue.

BUILT

6 · ONE FIX

Fixed once, for everyone

When a family or a type is wrong, it is corrected once, in the catalogue, and every job picks it up — the same rule Standard Details already runs on for details.

WITH STEP 2

WHAT WE ASK OF YOU

Engineers

- Mark beams from the **master beam schedule**. A new member is added to the schedule, not invented on a plan.
- A change of member or product goes through the schedule, so the mark, the schedule and the model move together.
- When the system asks you to confirm how a job is drafted, a one-line answer is enough.

Drafters

- Model the framing; don't draw it in 2D.
- Use the standard families once published. Don't load a family from another job or duplicate one to change it.
- If a type you need isn't there, ask for it to be added to the catalogue rather than making a local copy.

Measured 25 Oct 2026 from the drafted Revit models of 31207-01 and 31165-01 and the office wood master (read through an upgraded copy — the template itself was not changed), with the KOR Drafter tools and the KorStandards database. Counts: Sherwood 911 beams / 0 joist members / 212 span symbols; Edgar 2,085 beams in 7 families and 15 types, 5,459 joist members, 607 beam marks paired (564 right size, 197 another product, 177 of them B23), 2,366 of 7,544 framing members referenced to Level 1 from a higher storey. "No wood beam or joist family" in the standards template: no family named for beams, joists, lumber, TJI, Parallam or Timberstrand other than one steel HSS beam (KOR-Standards-Master-R25, read 25 Oct).